Low-Harmonic Push–Pull Class-E Power Amplifier With a Pair of *LC* Resonant Networks

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***Abstract*—This paper proposes a novel push–pull Class-E series-parallel resonant power amplifier (PA), which parallels two basic one-inductor, one-capacitor Class-E high-efficiency switching-mode tuned PA (Sokal, 1981) circuits. This proposed topology utilizes a push–pull scheme to realize sinusoidal output voltage. The previously published Class-E circuits (Sokal, 1981) can be implemented in a very simplified circuit. However, the simplified circuit is only appropriate for applications in which the harmonic content of output is not an important criterion. The proposed amplifier with two series-parallel resonant load networks will allow sinusoidal output voltage to be achieved by associating with the positive and negative quasi-sinusoidal waveforms. The complementarily activated configuration will provide continuous high-ripple-frequency input-current waveforms; this approach significantly reduces electromagnetic interference and requires very little filtering. With the symmetry of the push–pull Class-E circuit, there is the additional benefit that the even harmonics are suppressed at the load, and thus there are fewer harmonic distortions. However, each switch is symmetrically driven with a duty ratio of less than 0.5 to meet the nominal Class-E operating conditions, which leads to a narrower operating frequency variation. A prototype of the proposed PA shows over 95% efficiency at 48-V input, with 18-W output and an operating frequency of 900 kHz.**

***Index Terms*—Class-E power amplifier (PA), harmonic distortion (HD), high-efficiency amplifier, push–pull amplifier, switching PA, zero-voltage switching (ZVS).**

I. Introduction

In the high-efficiency Class-E power amplifier (PA) [2]–[9], the transistor is used as a switch. The resonator L_o, C_o is used to block the harmonic frequencies and dc component, forcing the output current I_o to approximate a sine wave at the fundamental frequency, with harmonic content as discussed in [10]. The radio frequency choke L_{RF} is assumed to be ideal such that it conducts only the dc current. The current into switch S and capacitor C_s must be a dc-offset sine wave, with some harmonic content as discussed in [10]. By appropriately adjusting the amplitude and phase of the load current, a solution is found with zero capacitor charge just prior to turn-on. This results in a switching waveform with zero voltage and zero voltage slope at turn-on. The conditions are those of the well-known *Class-E switching* [2], [11], [12]. This allows high-efficiency operation at frequencies up to 10 GHz, as of September 2005 [13]. Additionally, the Class-E topology can be implemented with fewer components because the power MOSFETs' parasitic capacitors can be incorporated into the circuit. These benefits have allowed the Class-E topology to achieve high power density, thus reducing the size and weight of the equipment. However, a blocking filter L_o, C_o is needed to block the harmonic frequencies and dc component, but also introduces loss [10], [14].

The shrinking size of electronic equipment demands ever-increasing power densities at high switching frequencies and a minimal parts count for the circuit technology. In an attempt to minimize the parts count with Class-E operation, the one-inductor one-capacitor Class-E high-efficiency switching-mode tuned PA [1] provides a more simplified circuit. Nevertheless, this simplified single-ended circuit is appropriate only for applications in which the harmonic content and the phase-modulation noise of the output are not important criteria. It is therefore desirable to retain the functions of the conventional Class-E features; i.e., that the amplifier can be operated with high efficiency at very high frequencies and provides a sinusoidal output waveform and power-handing capability without increasing the complexity of the power circuits.

Unlike the single-ended Class-E amplifier [1], the push–pull architecture is able to achieve a sinusoidal output waveform and high power-handing capability. For instance, a symmetrically driven push–pull Class-E amplifier has been proposed for high-power applications [15], as shown in Fig. 1(a). With the symmetrical gate-driving signals, theoretically, the even harmonics are entirely cancelled at the load, and thus there are fewer harmonic distortions (HDs). However, this doubled parts-count configuration incurs penalties on the overall efficiency and the design cost. Recently, the Class-E/F [16] and the current-mode Class-D [17], with low peak voltage and/or low rms current, have been implemented as a high-frequency amplifier, as shown in Fig. 1(b) and (c). However, the current-mode Class-D and the Class-E/F only achieve zero-voltage switching (ZVS) conditions. Fortunately, there is a more elegant way to further reduce the switching loss, if the switch current increases gradually from zero after the switch is closed.

This paper proposes a push–pull Class-E resonant PA with a simple *LC* load network and a load resistor R_L in each half-amplifier, as shown in Fig 1(d). An overlapped capacitor-voltage waveform is utilized to achieve the nominal Class-E conditions without increasing the complexity of the power circuits. For nominal operation, the following performance parameters are determined: the current and voltage waveforms, the peak values of drain current and drain-to-source voltage, the output power, the power-output capability, and the component values of the load network [18], [19]. The output total HD (THD) is also given to provide straightforward design function [20]. As a result of its simplicity, the design is suitable for dc–ac inverters, battery

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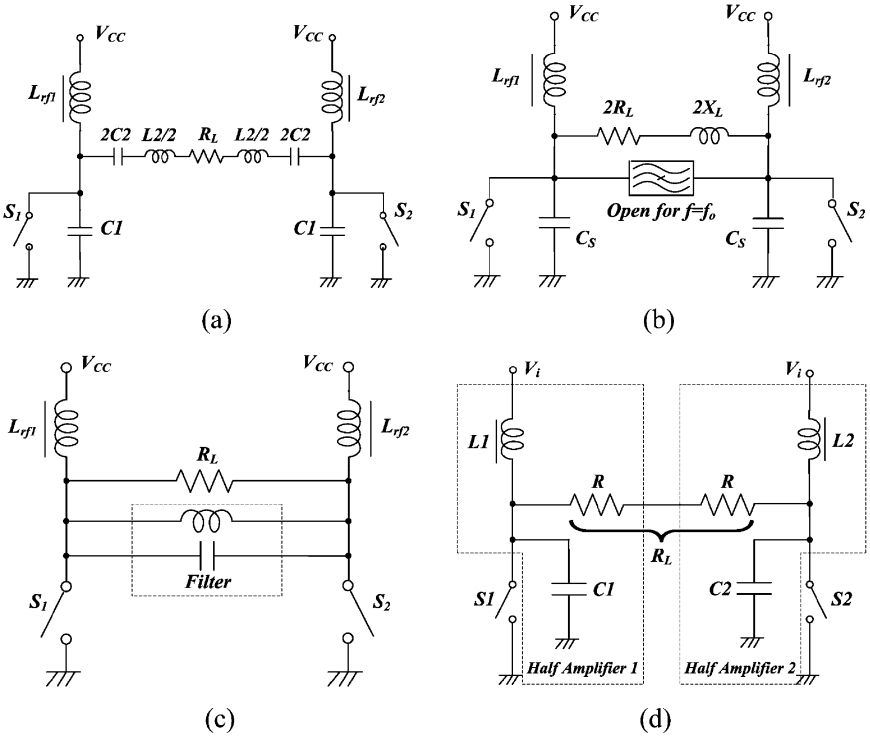


Fig. 1. Four classic push–pull amplifiers. (a) Dual-type original Class-E amplifier topology. (b) Class-E/F amplifier. (c) Class-D amplifier. (d) Proposed topology with a symmerical arrangement of one inductor and one capacitor in the load network.

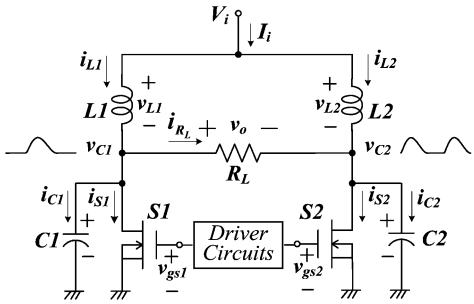


Fig. 2. Proposed push–pull Class-E PA.

chargers, induction heating, electronic ballasts, and other applications with similar demands [21]–[26]. Experiments carried out for a 48-V input, 18-W output push–pull Class-E PA, with a series-parallel *LCR* in load network, reveal an enhanced performance and support the preceding analysis

II. Principle of Operation

The basic schematic of the proposed push–pull Class-E series-parallel *LCR* resonant PA is shown in Fig. 2. It contains two transistors [these can be bipolar junction transistors (BJTs) or MOSFETs], two inductors, two capacitors, and a load resistance. Switches $S1$ and $S2$ are complementarily activated to drive periodically at the operating frequency $f = \omega/2\pi$ as in a push–pull switching PA [27], i.e., the switch waveforms are identical, except that the phase shifts between $S1$ and $S2$ are π with an "on" duty ratio D of less than 50%. The simplest type of half-amplifier, as shown in Fig. 1(d), is a series-parallel resonant circuit, which consists of an inductor L in series with a paralleled capacitor C and resistor R. The resistor R_L is the load to which the ac power is to be delivered, with neither end connected to a ground. It is suitable for a load that is balanced to a ground, but most RF-power loads have one end connected to a ground. To accommodate grounded loads, the proposed topology needs to add one of the following: a balun that can be used to provide the interface with the amplifier [28]; or a two-winding transformer (that has V_i connected to a center-tap on the primary winding), between the grounded load (on the grounded secondary winding) and the drains of $S1$ and $S2$ (connected to the ends of the center-tapped primary winding).

The switching sequences and theoretical waveforms for the steady-state operation of the proposed amplifier are illustrated in Fig. 3. To reduce the transistor turn-on power losses, the switch current i_S increases gradually from zero after the switch is closed. The proposed push–pull Class-E PA uses a pair of *LC* resonant networks with an overlapped capacitor-voltage waveform; this offers additional degrees of freedom, and thus there are two operational points that can validly achieve this situation:

Case 1) *[Zero-Voltage Zero-Slope Switching (ZVZSS)]*: In this case, the nominal operating conditions of ZVS and zero-voltage-slope switching (ZVSS) are simultaneously satisfied. Namely

$$\begin{cases} v_{C1}(\pi - 2\pi D) = 0 & (1) \\ \frac{dv_{C1}(\pi - 2\pi D)}{dt} = 0. & (2) \end{cases}$$

Case 2) *[Zero-Voltage Zero-Current Switching (ZVZCS)]*: The operation principle in the commutation of this case is solved by the following simultaneous equations:

$$\begin{cases} v_{C1}(\pi - 2\pi D) = 0 \\ i_{L1}(\pi - 2\pi D) = -\frac{v_{C2}(\pi - 2\pi D)}{R_L}. \end{cases} \quad (3)$$

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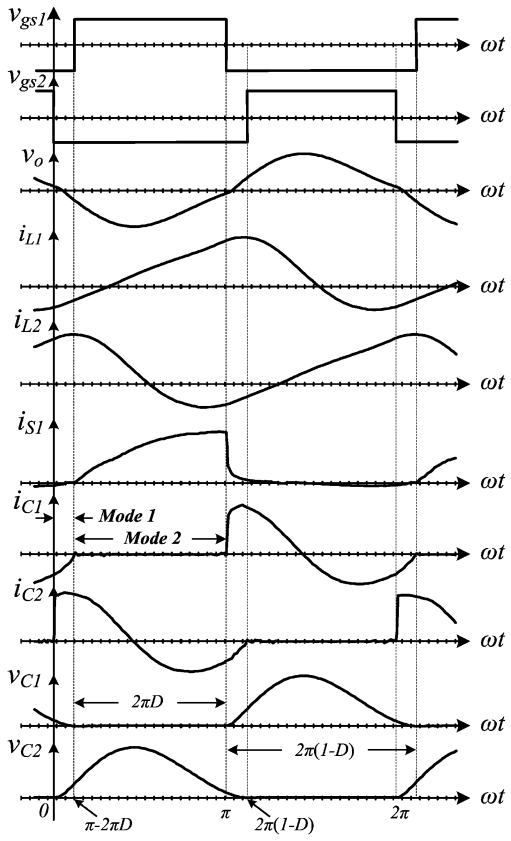


Fig. 3. Theoretical waveforms.

In order to satisfy both case 1 and case 2, it is necessary to find the current $i_{L1} = -i_{R_L}$ by which the switch current i_{S2} increases gradually from zero at time $t = (\pi - 2\pi D)/\omega$, as shown in Figs. 2 and 3. The duty ratio D must be kept at less than 50% so that the capacitor-voltage waveforms v_{C1} and v_{C2} can be overlapped. The formulas describing the waveforms are given in Section III-B.

III. Analysis of the Idealized Amplifier Circuit

A. Assumptions

The principles of operation in the steady-state condition are based on the idealized equivalent circuits shown in Fig. 4, under the following assumptions.

1) All the components and switches are ideal.
2) Inductances $L1$ and $L2$ are identical; $L1 = L2 = L$.
3) Capacitors $C1$ and $C2$ are identical; $C1 = C2 = C$. Any switch capacitance can be lumped into external capacitor C.

B. Steady-State Waveforms

The proposed amplifier produces the same half-sine resonant voltage at each switch's output as does the Class-E PA, but with a phase shift of π. Thus, only the half-switching period needs to be considered. Fig. 4 shows the topological equivalent circuits for a half-cycle of operation in the steady-state condition, in which each operating half-cycle can be divided into two

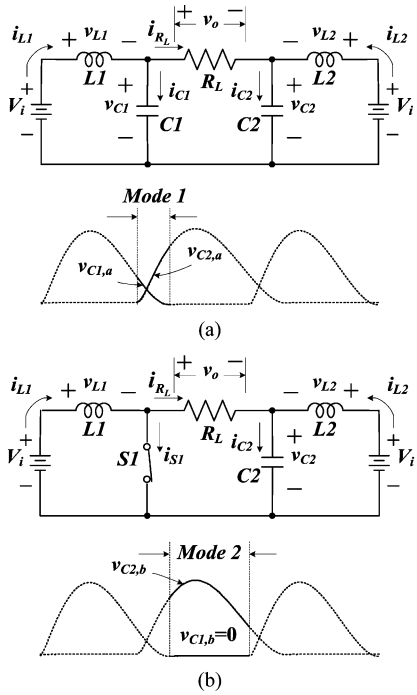


Fig. 4. Topological equivalent circuits in a half-cycle of operations. (a) Switches $S1$ and $S2$ both in the off-state. (b) Switch $S1$ in the on-state and $S2$ in the off-state.

modes [29]. The two modes of push–pull Class-E operation are described as follows.

1) Mode 1 $(0\sim\pi-2\pi D)$ *[*$S1, S2$ *off]:* In this range, switches $S1$ and $S2$ are both in the off-state, as shown in Fig. 4(a). The first step is to solve the state equations, which are derived in Appendix A. The results are as follows:

$$\frac{v_{C2,a}(\theta)}{V_i} = 1 + a_1\cos(A\theta) + a_2\sin(A\theta) + e^{-\frac{A\theta}{Q}}\left[a_3\cos\left(\frac{A\sqrt{Q^2-1}}{Q}\theta\right) + a_4\sin\left(\frac{A\sqrt{Q^2-1}}{Q}\theta\right)\right] \tag{4}$$

and as shown in (5) and (6), at the bottom of the next page, and as follows:

$$v_{C1,a}(\theta) = v_{C2,a}(\theta+\pi) \tag{7}$$

$$i_{L1,a}(\theta) = i_{L2,a}(\theta+\pi) \tag{8}$$

$$i_{C1,a}(\theta) = i_{C2,a}(\theta+\pi) \tag{9}$$

in which

$$Q = \omega_0 C R_L = \frac{R_L}{\omega_0 L} \tag{10}$$

$$A = \frac{\omega_0}{\omega} = \frac{f_0}{f} \tag{11}$$

where subscript a means the voltage (current) waveforms operation at Mode 1, $\omega_0 = 2\pi f_0 = 1/(LC)^{1/2}$ is the resonant frequency of the load network, $\omega = 2\pi f$ is an operating angular

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frequency, and $\theta = \omega t$ represents an angular displacement of the amplifier.

2) Mode 2 ($\pi - 2\pi D \sim \pi$) [$S1$ on, $S2$ off]: $S1$ commences turning on under nominal Class-E conditions at time $t = (\pi - 2\pi D)/\omega$, while switch $S2$ remains in the off-state. The equivalent circuit during this interval is shown in Fig. 4(b). In this range, inductor $L1$ is charged by the power source through power switch $S1$. Meanwhile, an inductor $L2$ in series with a paralleled capacitor $C2$ and resistor R_L series-parallel network is sustained for the resonant behavior of the previous state. A second-order differential equation is derived from Kirchhoff's laws; this state can be represented as

$$v''_{C2}(t) + \frac{1}{R_L \cdot C2} v'_{C2}(t) + \frac{1}{L2 \cdot C2} v_{C2}(t) = \frac{V_i}{L2 \cdot C2}. \tag{12}$$

The steady-state current and voltage waveforms are as shown in (13)–(17) at the bottom of the page, where subscript b means the voltage (current) waveforms operation at Mode 2. The output voltage is equal to the difference between v_{C1} and v_{C2}; thus

$$v_o(\theta) = v_{C1} - v_{C2} = \begin{cases} v_{C2,a}(\theta + \pi) - v_{C2,a}(\theta); & 0 < \theta \le \pi - 2\pi D \\ -v_{C2,b}(\theta - \pi + 2\pi D); & \pi - 2\pi D < \theta \le \pi \\ v_{C2,a}(\theta - \pi) - v_{C2,a}(\theta); & \pi < \theta \le 2\pi(1 - D) \\ v_{C2,b}(\theta - 2\pi + 2\pi D); & 2\pi(1 - D) < \theta \le 2\pi. \end{cases} \tag{18}$$

As the capacitor's voltage and inductor's current are always associated with energy, their values cannot suddenly change. Therefore, these values at the end of Mode 1 (Mode 2) should be the same as they are at the beginning of Mode 2 (Mode 1). Thus, the boundary and initial conditions are

$$\begin{cases} v_{C2,a}(0) = 0 \\ v_{C1,a}(0) = v_{C2,b}(\pi) \\ i_{L2,a}(0) = i_{L1,b}(\pi) \\ i_{L1,a}(0) = i_{L2,b}(\pi) \\ v_{C2,b}(\pi - 2\pi D) = v_{C2,a}(\pi - 2\pi D) \\ i_{L2,b}(\pi - 2\pi D) = i_{L2,a}(\pi - 2\pi D). \end{cases} \tag{19}$$

In addition, substituting nominal turn-on conditions (1), (2) and (3) into (4) yields other relationships among a_1, a_2, a_3, a_4, D, Q and A, and the coefficient a_1 gives (20)–(22), shown at the bottom of the next page, where $a_1 = A_1/V_i, a_2 = A_2/V_i, a_3 = A_3/V_i$ and $a_4 = A_4/V_i$ are the normalized coefficient of (A11).

The coefficients $a_1, a_2, a_3, a_4, b_1, b_2, D, Q$ and A can be solved numerically from the set of [see (19)–(21)] and [see (19), (20), (22)] for both case 1 (ZVZSS) and case 2 (ZVZCS), respectively, via a symbolic processor found in Mathematica 5.0 or higher versions (see Appendix B). The results are shown in Fig. 5, the calculated values for Q factor and ratio A are plotted together as a function of D at the two operating points for both case 1 (ZVZSS) and case 2 (ZVZCS).

$$\frac{i_{L2,a}(\theta) \cdot R_L}{V_i} = a_2 Q \cos(A\theta) - a_1 Q \sin(A\theta) + e^{-\frac{A\theta}{Q}} \left[\begin{array}{l} (a_3 + a_4\sqrt{Q^2-1}) \cos\left(\frac{A\sqrt{Q^2-1}}{Q}\theta\right) \\ +(a_4 - a_3\sqrt{Q^2-1}) \sin\left(\frac{A\sqrt{Q^2-1}}{Q}\theta\right) \end{array} \right] \tag{5}$$

$$\frac{i_{C2,a}(\theta) \cdot R_L}{V_i} = a_2 Q \cos(A\theta) - a_1 Q \sin(A\theta) + e^{-\frac{A\theta}{Q}} \left[\begin{array}{l} (-a_3 + a_4\sqrt{Q^2-1}) \cos\left(\frac{A\sqrt{Q^2-1}}{Q}\theta\right) \\ -(a_4 + a_3\sqrt{Q^2-1}) \sin\left(\frac{A\sqrt{Q^2-1}}{Q}\theta\right) \end{array} \right] \tag{6}$$

$$\frac{v_{C2,b}(\theta)}{V_i} = 1 + e^{-\frac{A(\theta - \pi + 2\pi D)}{2Q}} \left[\begin{array}{l} b_1 \cos\left(\frac{A\sqrt{4Q^2-1}}{2Q}(\theta - \pi + 2\pi D)\right) \\ +b_2 \sin\left(\frac{A\sqrt{4Q^2-1}}{2Q}(\theta - \pi + 2\pi D)\right) \end{array} \right] \tag{13}$$

$$\frac{i_{L2,b}(\theta) \cdot R_L}{V_i} = 1 + e^{\frac{-A(\theta - \pi + 2\pi D)}{2Q}} \left[\begin{array}{l} \frac{b_1 + b_2\sqrt{4Q^2-1}}{2} \cos\left(\frac{A\sqrt{4Q^2-1}}{2Q}(\theta - \pi + 2\pi D)\right) \\ +\frac{b_2 - b_1\sqrt{4Q^2-1}}{2} \sin\left(\frac{A\sqrt{4Q^2-1}}{2Q}(\theta - \pi + 2\pi D)\right) \end{array} \right] \tag{14}$$

$$\frac{i_{C2,b}(\theta) \cdot R_L}{V_i} = e^{\frac{-A(\theta - \pi + 2\pi D)}{2Q}} \left[\begin{array}{l} \frac{-b_1 + b_2\sqrt{4Q^2-1}}{2} \cos\left(\frac{A\sqrt{4Q^2-1}}{2Q}(\theta - \pi + 2\pi D)\right) \\ -\frac{b_2 + b_1\sqrt{4Q^2-1}}{2} \sin\left(\frac{A\sqrt{4Q^2-1}}{2Q}(\theta - \pi + 2\pi D)\right) \end{array} \right] \tag{15}$$

$$\frac{i_{L1,b}(\theta) \cdot R_L}{V_i} = QA(\theta - \pi + 2\pi D) + \frac{i_{L1,a}(\pi - 2\pi D) R_L}{V_i} \tag{16}$$

$$\frac{i_{S1,b}(\theta) \cdot R_L}{V_i} = QA(\theta - \pi + 2\pi D) + \frac{i_{L1,a}(\pi - 2\pi D) \cdot R_L}{V_i} + \frac{v_{C2,b}(\theta - \pi + 2\pi D)}{V_i} \tag{17}$$

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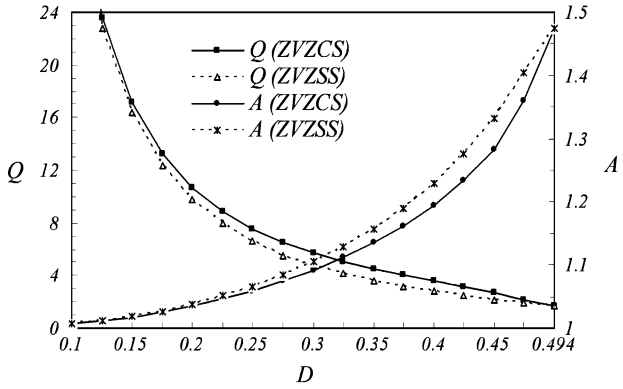


Fig. 5. Relationships among D, Q and A for both case 1 (ZVZSS) and case 2 (ZVZCS).

The maximum duty ratio $D_{\max}$ of the proposed push–pull Class-E amplifier exists such that (19)–(22) simultaneously hold true. Computed numerically, the solution in Mathematica is $D_{\max} = 0.4943353, Q_{\min} = 1.71533$ and $A = 1.47387$. The goal of the overlapped capacitor-voltage waveform is to satisfy ZVS and ZVSS at turn-on for duty ratio $D \leq D_{\max}$; otherwise only "suboptimum" conditions can be achieved. In the suboptimum topology, $D > D_{\max}$, a flyback diode across the switch (body diode) is used to damp the voltage across the capacitor, while $i_{C1}(\pi - 2\pi D) < 0$ results in a near-zero voltage across the switch. Transistor $S1$ is then turned on to achieve soft switching.

Fig. 5 reveals several interesting facts, listed as follows.

1) Since for the push–pull configuration to satisfy the nominal Class-E operating conditions the duty cycle is limited below 50%, the operating frequency variations of the proposed circuit in terms of ratio A are between 1 and 1.5, as shown in Fig. 5.
2) There are two operation points at the same duty ratio—for both ZVZSS and ZVZCS—that can be performed in order to satisfy the nominal Class-E operating conditions. It is shown that ZVZCS has a higher loaded Q-factor and lower ratio A than ZVZSS at the same duty ratio D. In other words, the ZVZCS switching provides higher output power with larger duty ratio when it operates at the same switching frequency.

The resistance that the amplifier offers to the dc power supply is

$$R_{\rm DC} = \frac{V_i}{I_I} = \frac{V_i}{\frac{1}{\pi}\left\{\begin{array}{l}\int_0^{\pi-2\pi D}[i_{L1,a}(\theta)+i_{L2,a}(\theta)]d\theta \\ +\int_{\pi-2\pi D}^{2\pi-2\pi D}[i_{L1,b}(\theta)+i_{L2,b}(\theta]d\theta\end{array}\right\}} \tag{23}$$

where the dc supply current I_I is equal to the average value of input current I_i. Fig. 6 shows the resistance ratio $R_{\rm DC}/R_L$ of the proposed amplifier as a function of D at the two operating points for both case 1 (ZVZSS) and case 2 (ZVZCS).

C. Peak Drain Current and Drain-to-Source Voltage

From (17), the peak drain current is

$$i_{S\max} = i_{S1,b}(\pi). \tag{24}$$

Finding that the derivative of (13) is equal to zero, the maximum value of the drain-to-source voltage can be expressed as (25), shown at the bottom of the next page.

$$a_1 = \sec[A(2\pi D - 2\pi)]\left\{-1 + a_2\sin[A(2\pi D - 2\pi)] + e^{\frac{A(2\pi D-2\pi)}{Q}}\left[\begin{array}{l}-a_3\cos\left(\frac{A\sqrt{Q^2-1}}{Q}(2\pi D - 2\pi)\right) \\ +a_4\sin\left(\frac{A\sqrt{Q^2-1}}{Q}(2\pi D - 2\pi)\right)\end{array}\right]\right\} \tag{20}$$

$$a_1 = \frac{-\sec[A(2\pi D - 2\pi)]}{Q}\left\{\begin{array}{l}a_2 Q\cos[A(2\pi D - 2\pi)] + e^{\frac{A(2\pi D-2\pi)}{Q}} \\ \left[\begin{array}{l}\left[(a_4\sqrt{Q^2-1} - a_3)\cos\left(\frac{A\sqrt{Q^2-1}}{Q}(2\pi D - 2\pi)\right)\right] \\ +(a_4 + a_3\sqrt{Q^2-1})\sin\left(\frac{A\sqrt{Q^2-1}}{Q}(2\pi D - 2\pi)\right)\end{array}\right]\end{array}\right\} \tag{21}$$

and

$$a_1 = \frac{\begin{array}{l}-1 - a_2 Q\cos[A(2\pi - 2\pi D)] - a_2\sin[A(\pi - 2\pi D)] \\ -e^{\frac{A(2\pi D-\pi)}{Q}}\left[\begin{array}{l}a_3\cos\left(\frac{A\sqrt{Q^2-1}}{Q}(\pi - 2\pi D)\right) \\ +a_4\sin\left(\frac{A\sqrt{Q^2-1}}{Q}(\pi - 2\pi D)\right)\end{array}\right] \\ -e^{\frac{A(2\pi D-2\pi)}{Q}}\left[\begin{array}{l}(a_3 + a_4\sqrt{Q^2-1})\cos\left(\frac{A\sqrt{Q^2-1}(2\pi-2\pi D)}{Q}\right) \\ +(a_4 - a_3\sqrt{Q^2-1})\sin\left(\frac{A\sqrt{Q^2-1}(2\pi-2\pi D)}{Q}\right)\end{array}\right]\end{array}}{\cos[A(\pi - 2\pi D)] - Q\sin[A(2\pi - 2\pi D)]} \tag{22}$$

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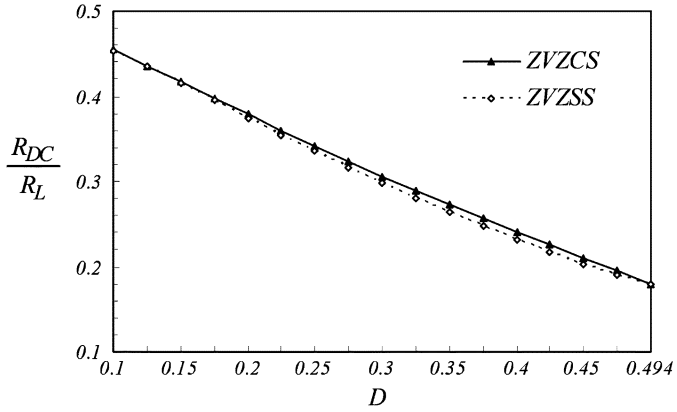


Fig. 6. Equivalent dc resistance as a function of D for both case 1 (ZVZSS) and case 2 (ZVZCS).

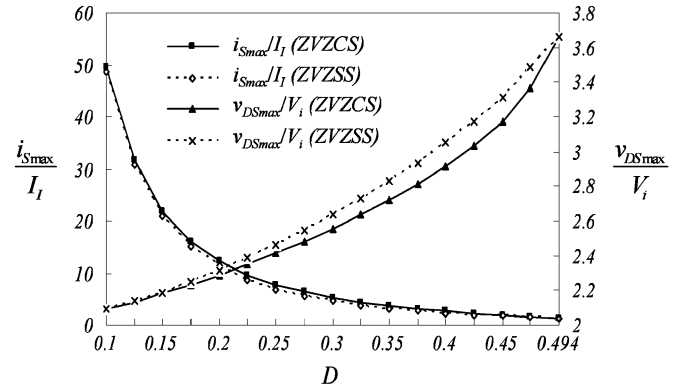


Fig. 7. Normalized peak drain current and drain-to-source voltage as a function of D for both case 1 (ZVZSS) and case 2 (ZVZCS).

In Fig. 7, the calculated normalized values for i_{Smax}/I_I and v_{DSmax}/V_i are shown together as a function of D at the two operating points for both case 1 (ZVZSS) and case 2 (ZVZCS).

D. Power Relationships

The input power is

$$P_{\mathrm{in}} = V_i I_I. \tag{26}$$

To simplify the calculation, the drain efficiency $\eta_C = 100\%$ is the Assumption 1. Therefore, output power P_o, consisting of the fundamental frequency component and all-harmonic frequency components, is equal to dc supply power P_{in}. The output power is approximately given by

$$P_o = \frac{V_i}{\pi}\left\{\begin{array}{c}\int_0^{\pi-2\pi D}[i_{L1,a}(\theta)+i_{L2,a}(\theta)]d\theta \\ +\int_{\pi-2\pi D}^{2\pi-2\pi D}[i_{L1,b}(\theta)+i_{L2,b}(\theta)]d\theta\end{array}\right\}. \tag{27}$$

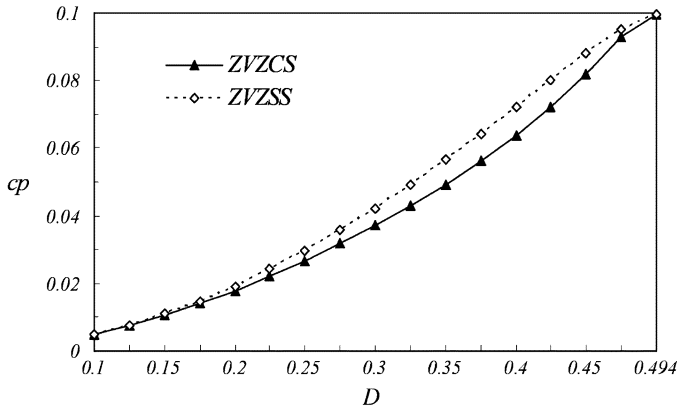


Fig. 8. The power-output capability as a function of D for both case 1 (ZVZSS) and case 2 (ZVZCS).

The power-output capability (per transistor) is defined as:

$$c_p = \frac{P_o/2}{i_{S\,\max}\cdot v_{\mathrm{DS\,max}}}. \tag{28}$$

Fig. 8 illustrates the output-power capability. Because the duty cycle must remain below 0.5, the full Class-E output-power capability of the transistors is never obtained.

E. Relationships Among Load-Network Design Element Values

Using (10) and (11), the relationships among load-network design element values are

$$\frac{\omega L}{R_L} = \frac{1}{AQ} \tag{29}$$

and

$$\omega C R_L = \frac{Q}{A}. \tag{30}$$

These results are shown in Fig. 9, and offer a simple design procedure for achieving nominal amplifier performance.

$$\frac{v_{\mathrm{DS\,max}}}{V_i} = 1+\frac{1}{2}\exp\frac{-\sec^{-1}\left\{\frac{2\sqrt{b_1^2+b_2^2}\left[-b_1^2+b_2^2(4Q^2-1)\right]}{\left[4b_1b_2Q^2+\left(b_1^2+b_2^2\right)\sqrt{4Q^2-1}\right]\sqrt{\frac{b_1^2-2b_1b_2\sqrt{4Q^2-1}+b_2^2(4Q^2-1)}{Q^2}}}\right\}}{\sqrt{4Q^2-1}}\left\{\begin{array}{c} b_2\sqrt{\frac{b_1^2-2b_1b_2\sqrt{4Q^2-1}+b_2^2(4Q^2-1)}{\left(b_1^2+b_2^2\right)Q^2}} \\ -\frac{b_1\left(\left[4b_1b_2Q^2+\left(b_1^2+b_2^2\right)\sqrt{4Q^2-1}\right]\sqrt{\frac{b_1^2-2b_1b_2\sqrt{4Q^2-1}+b_2^2(4Q^2-1)}{Q^2}}\right)}{\sqrt{b_1^2+b_2^2}\left[b_1^2-b_2^2(4Q^2-1)\right]}\end{array}\right\} \tag{25}$$

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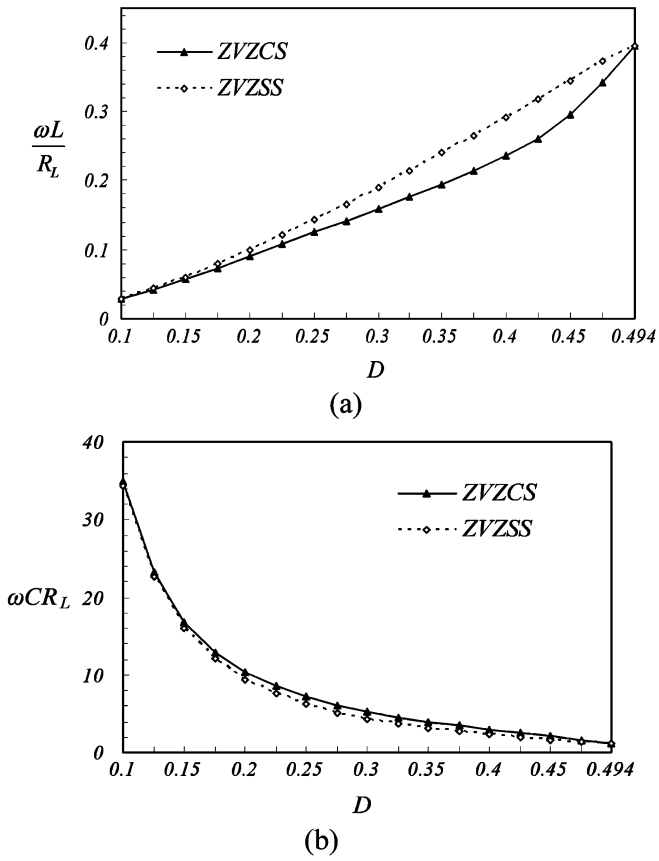


Fig. 9. Relationships among the element values of the load network; (a) $\omega L/R_L$; and (b) $\omega C R_L$, each is which is shown as a function of D for both case 1 (ZVZSS) and cases 2 (ZVZCS).

F. Fourier Series Expression and THD

The Fourier series expression can be readily developed for the *LCR* series-parallel resonant PA. In addition to being an important engineering design tool, frequency-domain characterization also plays an important role for analyzing the harmonic values in the resonant amplifier system, since the existence of these harmonics affects the overall efficiency, performance and cost of the power electronics system. The best tool available to study harmonics is Fourier analysis. The output-voltage waveform can be expressed as

$$v_o = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{n\pi\theta}{p}\right) + b_n \sin\left(\frac{n\pi\theta}{p}\right)\right] \tag{31}$$

in which

$$p = \frac{T}{2} \tag{32}$$

where a_0 is the average value, and a_n and b_n are the components at the nth harmonic frequency. The coefficients can be obtained from (4), (13), and (18) via a symbolic processor found in Mathematica 5.0 and higher versions (see Section B of the Appendix).

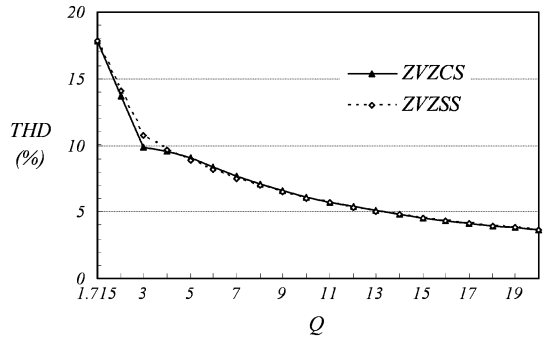


Fig. 10. THD as a function of Q for both case 1 (ZVZSS) and case 2 (ZVZCS).

TABLE I
EXPERIMENT PARAMETERS FOR THE AMPLIFIER SHOWN IN FIG. 2

	Fig. 12 (a) (*ZVZSS*)	Fig. 12 (b) (*ZVZCS*)
V_i (V):	48	48
S1, *S2*:	IRF630N $R_{DS(on)}$=0.267 Ω at V_{GS}=15V	IRF630N $R_{DS(on)}$=0.267 Ω at V_{GS}=15V
R_L (Ω):	798	798
$D \approx$	0.457	0.464
f (*kHz*)	917.367	906.574
L1, *L2* (μH):	28.2, 27.8	28.2, 27.8
C1, *C2* (*p*F):	527.8, 543.5	527.8, 543.5
I_i (A):	0.351	0.398
P_i (W):	16.848	18.72
P_o (W):	16.22	18.14
η (%):	96.27	96.9

The amount of distortion in the voltage or current waveform is quantified by means of an index called THD. The THD in the output voltage is defined as

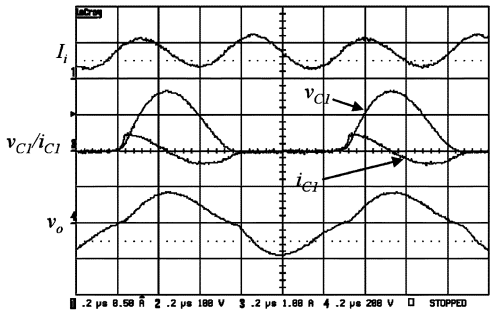
$$\mathrm{THD} = \frac{\sqrt{\sum_{n=2}^{\infty}(V_{\mathrm{on}})^2}}{V_{o1}} \tag{33}$$

where $V_{\mathrm{on}} = \sqrt{(a_n^2 + b_n^2)/2}$ is the rms value of the nth harmonic. The values of the THD are summed up to $n = 3000$ terms; Fig. 10 presents the THD variation of the output voltage as a function of Q at the two operating points for both case 1 and case 2. The THD increases with decreases in Q; the $\mathrm{THD}_{\max} = 17.827\%$ occurs at $Q_{\min} = 1.715$. This filter-less configuration allows that for a THD of less than 10%, Q values of greater than about 3 and 4 are needed for the operating points of the two cases. However, for the original Class-E amplifier, it is desirable to choose Q values greater than 7 [23].

IV. EXPERIMENTAL RESULTS

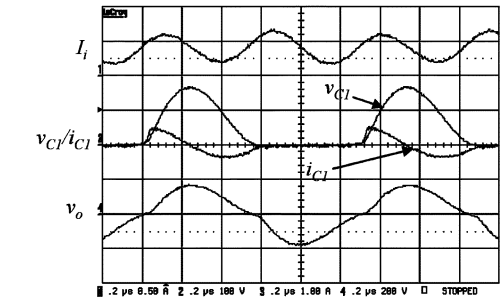
The prototype of the proposed push–pull Class-E series-parallel *LCR* resonant PA is designed with the specifications of 48 Vdc input and 18 W output to a load resistance $R_L = 798\ \Omega$. The parameters of both case 1 and case 2 for the PA shown in Fig. 2 are listed in Table I. Fig. 11(a) shows the waveforms for I_i, v_{C1}, i_{C1} and v_o in the circuit shown in Fig. 2 with duty

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I_i(0.5A/div) v_{C1}(100V/div) i_{C1}(1A/div) v_o(200V/div). Time Sscale (0.2 μs/div)

(a)



I_i(0.5A/div) v_{C1}(100V/div) i_{C1}(1A/div) v_o(200V/div). Time Scale (0.2 μs/div)

(b)

Fig. 11. Experimental waveforms for the amplifier shown in Fig. 2 with load resistance $R_L = 798\ \Omega$: (a) for duty ratio $D = 0.457$; and (b) for duty ratio $D = 0.464$, each of which is shown for both case 1 (ZVZSS) and case 2 (ZVZCS), respectively. (a) I_i (0.5 a/div) v_{cl} (100 V/div) i_{cl} (1 V/div) v_o (200 V/div). Time Sscale (0.2 μs/div) (b) I_i (0.5 a/div) v_{cl} (100 V/div) i_{cl} (1 V/div) v_o (200 V/div). Time Sscale (0.2 μs/div).

TABLE II
COMPARISON OF MEASURED HARMONIC CONTENTS OF OUTPUT VOLTAGE v_o

Harmonic of v_o	D=0.457 (*ZVZSS*)	
	Proposed	Conventional
2nd	0.74%	49.63%
3rd	11.85%	11.72%
4th	–	6.23%
5th	3.29%	3.37%
6th	–	2.49%
Total (rms)	**12.32%**	**51.54%**

ratio $D = 0.457$; these demonstrate that switch $S1$ achieves ZVS and ZVZSS conditions. Fig. 11(b) shows the waveforms for I_i, v_{C1}, i_{C1} and v_o in the circuit shown in Fig. 2 with duty ratio $D = 0.464$; these demonstrate that switch $S1$ achieves ZVS and ZVZSS conditions. The theoretical waveforms and parameters of the amplifier are well matched for normal experimental conditions. As predicted from Figs. 5 and 8, when both cases operate at the same loaded Q factor, the ZVZCS provides higher output power with larger duty ratio than does the ZVZSS case. The harmonic contents of output voltage are measured using the FFT function of a digital storage oscilloscope. Table II gives comparisons between the measured harmonic contents of the proposed push–pull Class-E amplifier and the conventional single-ended circuit configuration that includes one inductor and one capacitor [18]. As expected, the harmonic contents of output voltage are significantly reduced in the proposed push–pull amplifier. However, the amplitudes of the positive and negative half-cycle in the output-voltage waveform are not symmetrical, which may cause a small second-harmonic component, as shown in Fig. 11(a) and Table II, respectively. This result was not reached on the previous assumptions that the two transistors, inductors and capacitors are not identical (see Table I). Because of their nonlinearity and that the tolerance of the component characteristics differ appreciably, the appearance of even harmonics must be expected.

V. CONCLUSION

A novel push–pull Class-E series-parallel resonant PA was proposed and implemented for the case of the series-inductor, shunt-capacitor load-network configuration. The proposed amplifier achieves an overlapped capacitor-voltage waveform to meet the nominal Class-E conditions, in a manner similar to the Class-AB amplifier scheme that eliminates crossover distortion. In contrast to a single-ended configuration, the proposed push–pull amplifier accomplishes the following:

1) It provides a continuous, high-ripple-frequency input-current waveform.
2) Due to its symmetry and the duty ratio $D < 0.5$, the proposed amplifier achieves a significantly decreased level of THD ($< 17.827\%$).
3) The narrower operating frequency variation of ratio $A(f_o/f)$ changes to realize nominal amplifier performance for values higher than $Q_{\min}$.
4) The voltage and current stresses on the switches are lower than those of the original Class-E amplifier with the same output voltage and power.

The approaches presented here can be applied to the analysis and design of other Class-E amplifier configurations [18], or with more complicated circuits in exact designs. Further, it should be noted that for this topology, the circuit described in this paper has two operational points that are performed by the ZVZS and ZVZC switching. In future work the author plans to research how to control these operational points.

APPENDIX A
SOLVING THE STATE-SPACE EQUATIONS

When both switches $S1$ and $S2$ are off ($0 < \theta \leq \pi - 2\pi D$), a set of first-order differential equations are derived from Fig. 4(a) in this state as

$$\begin{cases} L\frac{di_{L1}}{dt} = -v_{c1} + V_i \\ L\frac{di_{L2}}{dt} = -v_{c2} + V_i \\ C\frac{dv_{c1}}{dt} = i_{L1} - \frac{1}{R_L}v_{c1} + \frac{1}{R_L}v_{c2} \\ C\frac{dv_{c2}}{dt} = i_{L2} - \frac{1}{R_L}v_{c1} + \frac{1}{R_L}v_{c2}. \end{cases} \tag{A1}$$

The matrix format is

$$\frac{d\mathbf{y}(t)}{dt} = \mathbf{A}\mathbf{y}(t) + \mathbf{B}V_i \tag{A2}$$

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where

$$\mathbf{y}(t) = [i_{L1}(t) \quad i_{L2}(t) \quad v_{C1}(t) \quad v_{C2}(t)]^T \tag{A3}$$

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & \frac{-1}{L} & 0 \\ 0 & 0 & 0 & \frac{-1}{L} \\ \frac{1}{C} & 0 & \frac{-1}{R_L C} & \frac{1}{R_L C} \\ 0 & \frac{1}{C} & \frac{1}{R_L C} & \frac{-1}{R_L C} \end{bmatrix} \tag{A4}$$

and

$$\mathbf{B} = \begin{bmatrix} \frac{1}{L} & \frac{1}{L} & 0 & 0 \end{bmatrix}^T. \tag{A5}$$

Let the general solution of a state-space equation is

$$\mathbf{y}(t) = \mathbf{P}\mathbf{z}(t) \tag{A6}$$

then

$$\dot{\mathbf{y}}(t) = \mathbf{P}\dot{\mathbf{z}}(t) = \mathbf{A}\mathbf{P}\mathbf{z}(t) + \mathbf{B}V_i \tag{A7}$$

$$\dot{\mathbf{z}}(t) = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}\mathbf{z}(t) + \mathbf{P}^{-1}\mathbf{B}V_i \tag{A8}$$

where $\mathbf{z}(t) = [z_1(t)z_2(t)z_3(t)z_4(t)]^T$, $\mathbf{P}$ is the eigenvectors of $\mathbf{A}$ and $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ is the diagonal matrix has the eigenvalues of $\mathbf{A}$ as its main diagonal elements. By finding the eigenvalues and eigenvectors of $\mathbf{A}$, the (A8) can easily be solved with Mathematica algorithm, thus

```
Eigenvalues[A]//MatrixForm
Eigenvectors[A]//MatrixForm
Transpose[%]//MatrixForm
Inverse[%]//MatrixForm]
```
(A9)

and

$$\begin{aligned}
&\texttt{DSolve}[\texttt{z1}'[\texttt{t}] == -\frac{\texttt{i}}{\sqrt{\texttt{C}}\sqrt{\texttt{L}}}\texttt{z1}[\texttt{t}] + \frac{\texttt{i}}{2\sqrt{\texttt{C}}\sqrt{\texttt{L}}}\texttt{Vi}, \texttt{z1}[\texttt{t}], \texttt{t}] \\
&\texttt{DSolve}[\texttt{z2}'[\texttt{t}] == \frac{\texttt{i}}{\sqrt{\texttt{C}}\sqrt{\texttt{L}}}\texttt{z2}[\texttt{t}] - \frac{\texttt{i}}{2\sqrt{\texttt{C}}\sqrt{\texttt{L}}}\texttt{Vi}, \texttt{z2}[\texttt{t}], \texttt{t}] \\
&\texttt{DSolve}[\texttt{z}'3[\texttt{t}] == \frac{-\texttt{L} - \sqrt{\texttt{L}^2 - \texttt{CL}{R_L}^2}}{\texttt{CL}R_L}\texttt{z3}[\texttt{t}], \texttt{z3}[\texttt{t}], \texttt{t}] \\
&\texttt{DSolve}[\texttt{z}'4[\texttt{t}] == \frac{-L + \sqrt{L^2 - \texttt{CL}{R_L}^2}}{\texttt{CL}R_L}\texttt{z4}[\texttt{t}], \texttt{z4}[\texttt{t}], \texttt{t}].
\end{aligned} \tag{A10}$$

Substituting the results of (A9) and (A10) into (A6), the voltage v_{C2} can now be expressed in the following form:

$$\begin{aligned}
v_{C2}(t) &= V_i + A_1 \cos \texttt{w}_0 t + A_2 \sin \texttt{w}_0 t \\
&\quad + e^{-\alpha_1 t}(A_3 \cos \texttt{w}_{N1} t + A_4 \sin \texttt{w}_{N1} t) \\
&= V_i \begin{bmatrix} 1 + a_1 \cos \texttt{w}_0 t + a_2 \sin \texttt{w}_0 t \\ + e^{-\alpha_1 t}(a_3 \cos \texttt{w}_{N1} t + a_4 \sin \texttt{w}_{N1} t) \end{bmatrix}
\end{aligned} \tag{A11}$$

where

$$\begin{aligned}
\alpha_1 &= \frac{1}{R_L \cdot C} = \frac{\omega_0}{Q} \\
\omega_{N1} &= \sqrt{\omega_0^2 - \alpha_1^2} = \omega_0 \sqrt{1 - \frac{1}{Q^2}} \\
A_1 &= a_1 \cdot V_i \\
A_2 &= a_2 \cdot V_i \\
A_3 &= a_3 \cdot V_i \\
A_4 &= a_4 \cdot V_i.
\end{aligned} \tag{A12}$$

APPENDIX B
PROOF OF (20), (21), (22) AND THE COEFFICIENTS $a_1, a_2, a_3, a_4, b_1, b_2, D, Q, A, a_n, b_n$

1) Proof of (20) with Mathematica Algorithm

$$\texttt{Solve}[\texttt{vC2a}[2\pi - 2\pi\texttt{d}] == 0, \texttt{a1}]. \tag{A13}$$

2) Proof of (21)

$$\texttt{Solve}[\texttt{iC2a}[2\pi - 2\pi\texttt{d}] == 0, \texttt{a1}]. \tag{A14}$$

3) Proof of (22)

$$\texttt{Solve}[\texttt{iL2a}[2\pi - 2\pi\texttt{d}] == \frac{-\texttt{vC2a}[\pi - 2\pi\texttt{d}]}{1}, \texttt{a1}]. \tag{A15}$$

4) Evaluation of the Coefficients a_1, a_2, a_3, a_4, b_1 and b_2

$$\begin{aligned}
&\texttt{Solve}[\{\texttt{vC2a}[0] == 0, \texttt{vC1a}[0] == \texttt{vC2b}[2\pi\texttt{d}] \\
&\texttt{iL2a}[0] == \texttt{iL1b}[2\pi\texttt{d}], \texttt{iL1a}[0] == \texttt{iL2b}[2\pi\texttt{d}] \\
&\texttt{vC2b}[0] == \texttt{vC2a}[\pi - 2\pi\texttt{d}] \\
&\texttt{iL2b}[0] == \texttt{iL2a}[\pi - 2\pi\texttt{d}]\}, \{a1, a2, a3, a4, b1, b2\}].
\end{aligned} \tag{A16}$$

5) Evaluation of the coefficients D, Q, and $A$with $D_{\max} = 0.494$ for Case 1

$$\begin{aligned}
&\texttt{d} = 0.4943 \\
&\texttt{FindRoot}\,[\{\texttt{a1} == \texttt{Sec}[2\texttt{A}(\texttt{d}-1)\pi]\,(\texttt{a2Sin}[2\texttt{A}(\texttt{d}-1)\pi] \\
&\quad -1 + e^{\frac{2A(d-1)\pi}{Q}}\left(\texttt{a4Sin}\left[\frac{2\texttt{A}(\texttt{d}-1)\pi\sqrt{\texttt{Q}^2-1}}{\texttt{Q}}\right]\right. \\
&\quad \left.-\texttt{a3Cos}[\frac{2\texttt{A}(\texttt{d}-1)\pi\sqrt{\texttt{Q}^2-1}}{\texttt{Q}}]\right)\Big) \\
&\texttt{a1} == \frac{-\texttt{Csc}[2\texttt{A}(\texttt{d}-1)\pi]}{\texttt{Q}}\,(\texttt{a2QCos}[2\texttt{A}(\texttt{d}-1)\pi] \\
&\quad + \texttt{e}^{\frac{2A(d-1)\pi}{Q}}\left((\texttt{a4}\sqrt{\texttt{Q}^2-1} - \texttt{a3})\texttt{Cos}[\frac{2\texttt{A}(\texttt{d}-1)\pi\sqrt{\texttt{Q}^2-1}}{\texttt{Q}}]\right. \\
&\quad \left.+(\texttt{a3}\sqrt{\texttt{Q}^2-1} + \texttt{a4})\texttt{Sin}\left[\frac{2\texttt{A}(\texttt{d}-1)\pi\sqrt{\texttt{Q}^2-1}}{\texttt{Q}}\right]\right)\Big)\Big\} \\
&\{\texttt{Q}, 1.715\}, \{\texttt{A}, 1.474\}].
\end{aligned} \tag{A17}$$

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6) Evaluation of the Fourier coefficients a_n and b_n

$$\begin{aligned}
\mathtt{an[n_]} = \frac{1}{\pi}\Bigg(& \int_0^{(1-2\mathrm{d})\pi} (\mathtt{vC2a[}\theta\mathtt{]} - \mathtt{vC2a[}\theta+\pi\mathtt{]})\mathtt{Cos[n}\theta\mathtt{]d}\theta \\
& + \int_{(1-2\mathrm{d})\pi}^{\pi} \mathtt{vC2b[}\theta - \pi(1-2\mathrm{d})\mathtt{]Cos[n}\theta\mathtt{]d}\theta \\
& + \int_{\pi}^{2(1-d)\pi} (\mathtt{vC2a[}\theta\mathtt{]} - \mathtt{vC2a[}\theta-\pi\mathtt{]})\mathtt{Cos[n}\theta\mathtt{]d}\theta \\
& + \int_{2(1-d)\pi}^{2\pi} (-\mathtt{vC2b[}\theta - 2\pi(1-\mathrm{d})\mathtt{]})\mathtt{Cos[n}\theta\mathtt{]d}\theta\Bigg) \\
\mathtt{bn[n_]} = \frac{1}{\pi}\Bigg(& \int_0^{(1-2\mathrm{d})\pi} (\mathtt{vC2a[}\theta\mathtt{]} - \mathtt{vC2a[}\theta+\pi\mathtt{]})\mathtt{Sin[n}\theta\mathtt{]d}\theta \\
& + \int_{(1-2\mathrm{d})\pi}^{\pi} \mathtt{vC2b[}\theta - \pi(1-2\mathrm{d})\mathtt{]Sin[n}\theta\mathtt{]d}\theta \\
& + \int_{\pi}^{2(1-\mathrm{d})\pi} (\mathtt{vC2a[}\theta\mathtt{]} - \mathtt{vC2a[}\theta-\pi\mathtt{]})\mathtt{Sin[n}\theta\mathtt{]d}\theta \\
& + \int_{2(1-\mathrm{d})\pi}^{2\pi} (-\mathtt{vC2b[}\theta - 2\pi(1-\mathrm{d})\mathtt{]})\mathtt{Sin[n}\theta\mathtt{]d}\theta\Bigg).
\end{aligned} \tag{A18}$$

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